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TRANSPORT PHENOMENA II, CHE 311

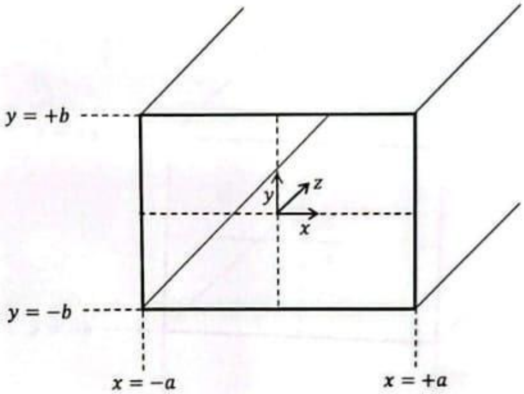
QUIZ 3

CLOSED BOOK, TIME: 10 MINUTES

A fluid with constant density ρ and viscosity μ flows through a rectangular duct of length L. The modified pressure at the inlet and the outlet of the duct are $\mathcal{P}_0$ and $\mathcal{P}_L$.

1- Write the partial differential equation (PDE) that describes the steady-state velocity profile of the fluid.

2- Write four boundary conditions needed to solve the above PDE.



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THE EQUATION OF MOTION FOR A NEWTONIAN FLUID WITH CONSTANT ρ AND μ

$$[\rho D\mathbf{v}/Dt = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g}]$$

Cartesian coordinates (x,y,z):

$$\rho\left(\frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}\right) = -\frac{\partial p}{\partial x} + \mu\left[\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2}\right] + \rho g_x \quad \text{(B.6-1)}$$

$$\rho\left(\frac{\partial v_y}{\partial t} + v_x\frac{\partial v_y}{\partial x} + v_y\frac{\partial v_y}{\partial y} + v_z\frac{\partial v_y}{\partial z}\right) = -\frac{\partial p}{\partial y} + \mu\left[\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2}\right] + \rho g_y \quad \text{(B.6-2)}$$

$$\rho\left(\frac{\partial v_z}{\partial t} + v_x\frac{\partial v_z}{\partial x} + v_y\frac{\partial v_z}{\partial y} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial p}{\partial z} + \mu\left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2}\right] + \rho g_z \quad \text{(B.6-3)}$$

$V_x = V_y = 0$

$V_z = V_z(x, y)$

$$0 = -\frac{\partial P}{\partial z} + \mu\left[\frac{\partial^2 V_z}{\partial x^2} + \frac{\partial^2 V_z}{\partial y^2}\right] + \rho g_z$$

$$P_L - P_0 = \nu\left[\frac{\partial^2 V_z}{\partial x^2} + \frac{\partial^2 V_z}{\partial y^2}\right]$$

no slip boundaries B.C.

$$\begin{cases} V_z(x=-a, y) = 0 \\ V_z(x=a, y) = 0 \\ V_z(x, y=-b) = 0 \\ V_z(x, y=+b) = 0 \end{cases}$$